

Market Share Dynamics Under Brand Switching

A Markov-Chain Simulation of Loyalty, Advertising, and Coupon Strategy

Portfolio Project Report

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*Excel-based stochastic simulation | 10,000 customers | 52-week horizon
Source: MGT239 Simulation for Business Course Project*

Executive Summary

This project examines **how brand loyalty and customer switching shape market share over time, and whether two marketing interventions—advertising and periodic coupons—create economic value in a stylized Coke-versus-Pepsi regional market.** The model treats weekly brand choice as a **two-state Markov process**. Under the baseline transition matrix, a Coke customer remains with Coke with probability 0.90, while a Pepsi customer remains with Pepsi with probability 0.80. The market contains 10,000 customers, each making one weekly purchase, and the simulation follows the system for 52 weeks.

Two stochastic implementations are used. The **macro model** simulates aggregate Coke and Pepsi customer counts directly through binomial transitions; the **micro model** simulates 10,000 individual customer purchase paths and then aggregates them into market shares. The theoretical baseline equilibrium for Coke is 66.67%. In the workbook's current realization, the macro model ends Week 52 at **66.70% Coke share** and the micro model at **66.10%**, providing a close cross-model check. Three initial conditions—50%, 10%, and 90% Coke share—end at 66.70%, 66.69%, and 66.75%, respectively, showing that the **starting state strongly affects the transition path but not the long-run region under a fixed transition matrix.**

The marketing experiments change the transition structure. **Advertising** raises Coke retention from 90% to 95% at an annual cost of \$50,000. In the simulated year, annual profit increases from \$341,209 to \$354,243 after advertising cost, a gain of \$13,034 (3.82%). A periodic \$0.25 **coupon** produces annual profit of \$357,852.75, a gain of \$16,643.75 (4.88%) over baseline. In this realization, **the coupon strategy produces \$3,609.75 more annual profit than advertising.** The managerial lesson is not that market share should be maximized mechanically, but that **changes in retention and switching must be evaluated through their dynamic profit consequences.**

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1. Business Problem and Research Questions

Market share is an outcome of repeated customer decisions. A firm can gain share because existing customers remain loyal, because customers switch from a competitor, or because marketing actions alter those transition probabilities. The central business difficulty is that a higher market share is not automatically equivalent to a higher profit. **Advertising** requires spending, and **coupons** intentionally reduce current margin. The relevant question is therefore dynamic: **do the future customer-state changes created by an intervention generate enough incremental contribution to justify its cost?**

The simulated setting is a regional cola market with two competing brands, **Coke and Pepsi**. The market is fixed at **10,000 customers**. Each customer buys one case per week and is classified by the brand purchased in the previous period. This structure makes the customer's current brand the state variable for predicting the next purchase. The project addresses four connected questions.

- **Q1 — Baseline market dynamics.** How does Coke's market share evolve over one year under the baseline loyalty and switching probabilities?
- **Q2 — Initial-share sensitivity.** If Coke starts with 10%, 50%, or 90% share, do the scenarios converge to a common long-run market-share region?
- **Q3 — Advertising decision.** If advertising reduces Coke-to-Pepsi switching from 10% to 5% but costs \$50,000 per year, does the resulting retention improvement increase annual profit?
- **Q4 — Coupon promotion.** If a \$0.25 coupon is issued periodically and temporarily changes switching behavior, does the induced share gain compensate for the margin sacrifice?

These questions connect customer behavior to managerial action: the first two establish the market's endogenous dynamics, while the latter two test whether changing those dynamics creates economic value.

2. Theoretical Framework

2.1 Brand Switching as a Markov Process

A **first-order Markov model** assumes that the probability of the next brand choice depends on the **current brand state**. This is a natural representation for repeat-purchase markets in which loyalty and switching are central. Prior marketing research has explicitly connected individual Markovian brand switching to aggregate market-share behavior (Givon & Horsky, 1994), while spectral approaches show how a stationary first-order transition matrix determines both the steady state and the speed of convergence toward it (Grover & Dillon, 1988).

The baseline transition matrix in the workbook is:

Current brand	Next: Coke	Next: Pepsi
Coke	0.90	0.10
Pepsi	0.20	0.80

The rows describe the customer's current brand and the columns describe the next purchase. Thus, 90% of Coke customers remain with Coke in the next week, while 10% switch to Pepsi. Among Pepsi customers, 80% remain with Pepsi and 20% switch to Coke.

Let $S_t = [s_{C,t}, s_{P,t}]$ denote the market-share row vector at week t . In expectation, the market evolves according to $S_{t+1} = S_t P$. A stationary share vector S^* satisfies $S^* = S^* P$. For Coke, the equilibrium condition is $s = 0.90s + 0.20(1-s)$, which gives $s = 2/3 = 66.67\%$. The equilibrium is therefore a property of the transition probabilities, not of the initial market share.

2.2 Why Simulation?

The analytical steady state provides a useful benchmark, but the workbook intentionally models realized customer counts as stochastic. A finite market does not move exactly along its expected path: each week contains random variation in how many customers remain loyal and how many switch. **Spreadsheet Monte Carlo methods** are well suited to such business models because random functions and inverse distribution functions can be combined directly with operational logic (Leong, 2007). More generally, simulation supports decision analysis by propagating uncertainty through a model and comparing the resulting outcomes of alternative decisions (Bielza, Müller, & Ríos Insua, 1999).

The simulation therefore serves two purposes. First, it visualizes the **transition path** and the **finite-sample noise** around the theoretical equilibrium. Second, it provides a common **computational framework for interventions** whose costs and time-varying transition probabilities make a purely static market-share comparison inadequate.

3. Model and Simulation Design

3.1 Core Inputs and Assumptions

Input	Workbook value	Role in model
Market size	10,000 customers	Fixed customer population
Horizon	52 weeks	One simulated year
Baseline Coke retention	0.90	$P(\text{Coke next} \mid \text{Coke current})$
Baseline Pepsi retention	0.80	$P(\text{Pepsi next} \mid \text{Pepsi current})$
Profit per case	\$1.00	Baseline weekly contribution per Coke purchase
Advertising Coke retention	0.95	Retention after advertising
Advertising annual cost	\$50,000	Deducted from annual profit
Coupon discount	\$0.25	Margin reduction in coupon weeks
Coupon-week Coke retention	0.95	Coke loyalty during promotion
Coupon-week Pepsi retention	0.60	Implies 40% Pepsi-to-Coke switching

The model assumes a closed **two-brand market**, one purchase per customer per week, and homogeneous transition probabilities within each current-brand state. The marketing experiments operate by **changing transition probabilities** rather than by adding an independent demand shock.

3.2 Macro Simulation

The **macro model** tracks the system state directly as customer counts. Let C_t be the number of Coke customers and $I_t = N - C_t$ the number of Pepsi customers. Conditional on the current state, next week's Coke customers come from two stochastic sources: Coke customers who remain loyal and Pepsi customers who switch to Coke. The workbook implements this logic with binomial random draws:

$$C_{t+1} = \text{Binomial}(C_t, p_{\text{Coke}}) + \text{Binomial}(N - C_t, 1 - p_{\text{Pepsi}}).$$

Coke market share is then C_t/N . This implementation is computationally compact because it simulates the **aggregate number** of transitions rather than every individual customer's path.

3.3 Micro Simulation

The micro model represents the same behavioral process from the bottom up. Each of 10,000 rows is a **customer** and each of 52 weekly columns records a binary brand state: 0 for Coke and 1 for Pepsi. The next state is generated conditionally from the current state using the appropriate transition probability. Weekly market shares are then obtained by **aggregating** the 10,000 individual states.

Using both implementations provides an internal consistency check. The macro and micro models use different computational granularity but represent the same transition system; therefore, their aggregate market-share paths should be close, subject to independent random variation.

3.4 Experimental Design

The workbook uses **four experiments**. **Q1** establishes the **baseline 52-week stochastic path** from a 50/50 market. **Q2** changes only the **initial market share**, comparing 50/50, 10/90, and 90/10 starting conditions. **Q3** changes Coke retention to 95% and deducts the \$50,000 **advertising cost**. **Q4** introduces a \$0.25 **coupon** on the workbook's promotion schedule (weeks 1, 6, 11, ..., 51); in those weeks the transition matrix becomes $[[0.95, 0.05], [0.40, 0.60]]$. All reported numerical results below are taken from the cached outputs in the final Excel workbook.

4. Results

4.1 Baseline Dynamics and Macro–Micro Agreement

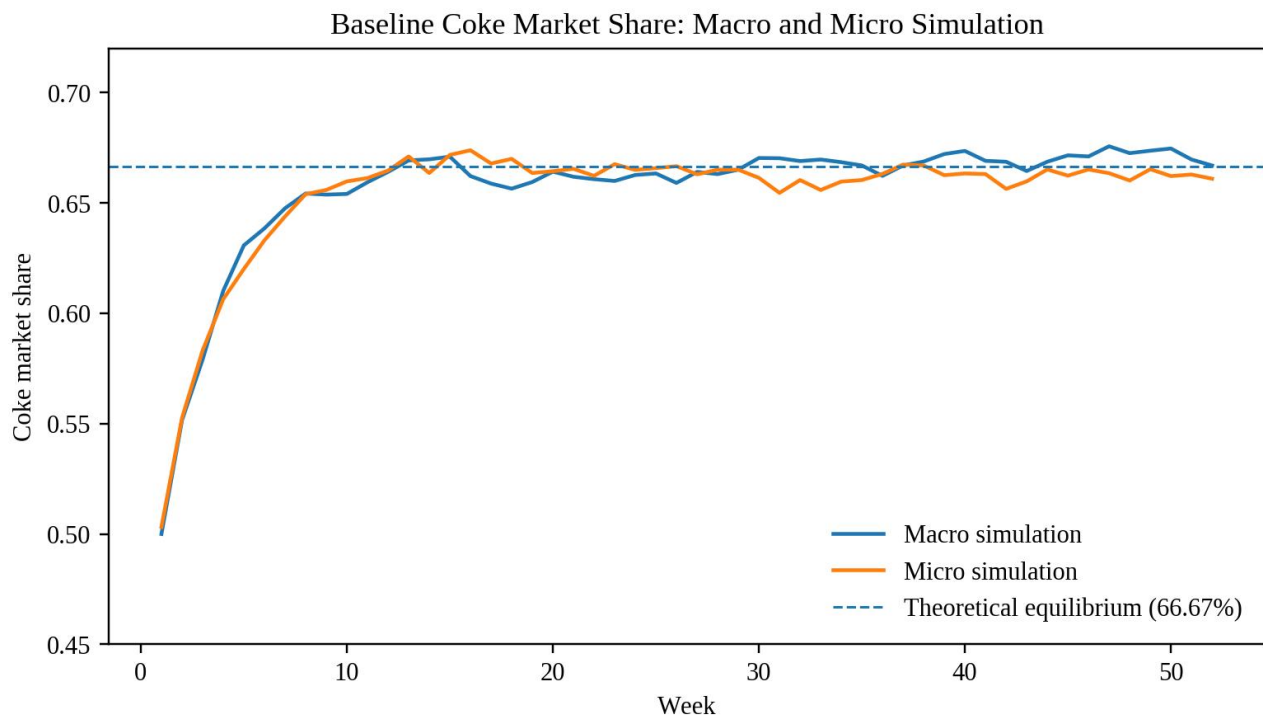


Figure 1. Baseline Coke market share under macro and micro simulation. Dashed line shows the 66.67% theoretical equilibrium.

The baseline **macro** simulation begins at 50% Coke share and moves rapidly toward the theoretical equilibrium. By Week 52, Coke share is **66.70%**. The **micro** simulation ends at **66.10%**. Across all 52 weeks, the mean absolute difference between the two simulated Coke-share paths is 0.58%, and the largest weekly absolute difference is 1.57%. The two implementations therefore tell the same substantive story while preserving the stochastic variation expected from independent random draws.

The **average Coke share** across the 52-week baseline **macro** path is **65.62%**. This annual average is below the long-run equilibrium because the simulation begins at 50% and spends the early weeks converging upward. The final-week value is a better diagnostic of long-run convergence than the full-year average in this particular initialization.

4.2 Initial Conditions and Convergence

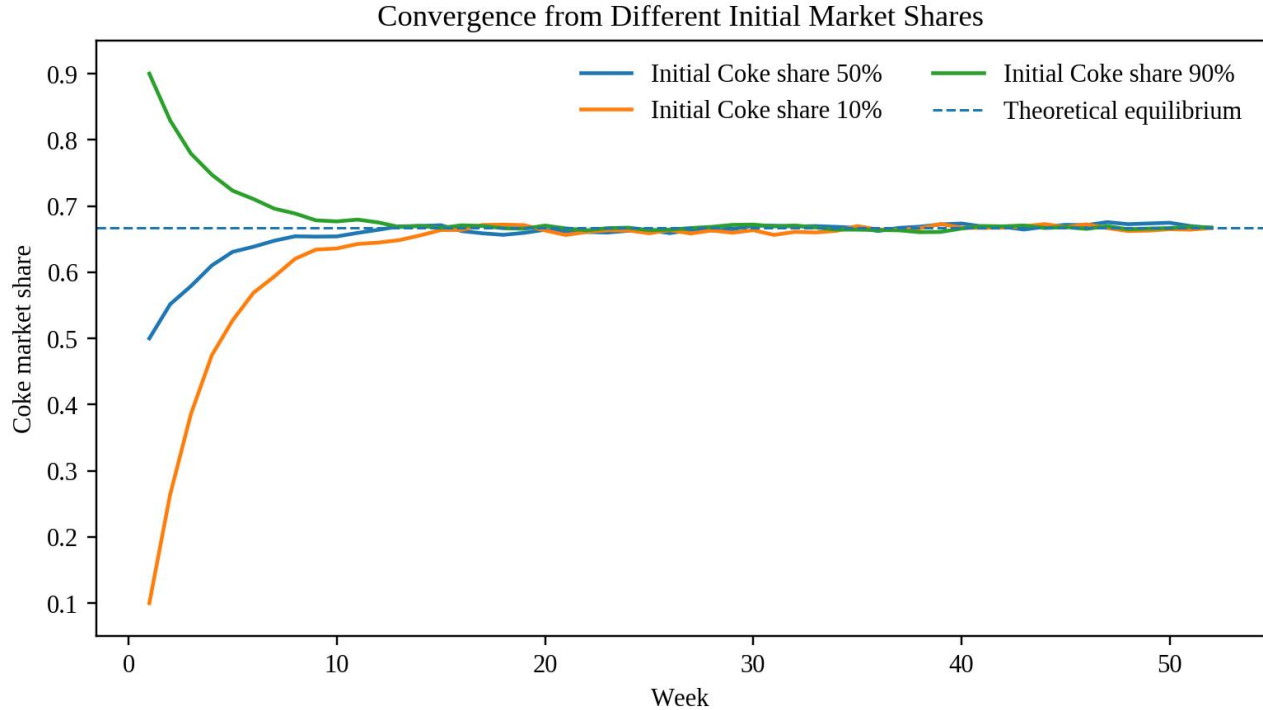


Figure 2. Coke market-share convergence from three initial conditions: 50%, 10%, and 90%.

Initial Coke share	Week 52 Coke share	Absolute gap from 66.67% equilibrium
50%	66.70%	0.033%
10%	66.69%	0.023%
90%	66.75%	0.083%

The three scenarios exhibit very different early trajectories but **converge tightly** by the end of the horizon. Starting Coke shares of **50%**, **10%**, and **90%** finish at **66.70%**, **66.69%**, and **66.75%**, respectively. Their Week-52 deviations from the theoretical 66.67% equilibrium are only 0.033, 0.023, and 0.083 percentage points. The result demonstrates that **the initial condition governs the transition path but has little influence on the long-run region when the transition matrix is fixed and ergodic.**

4.3 Advertising Experiment

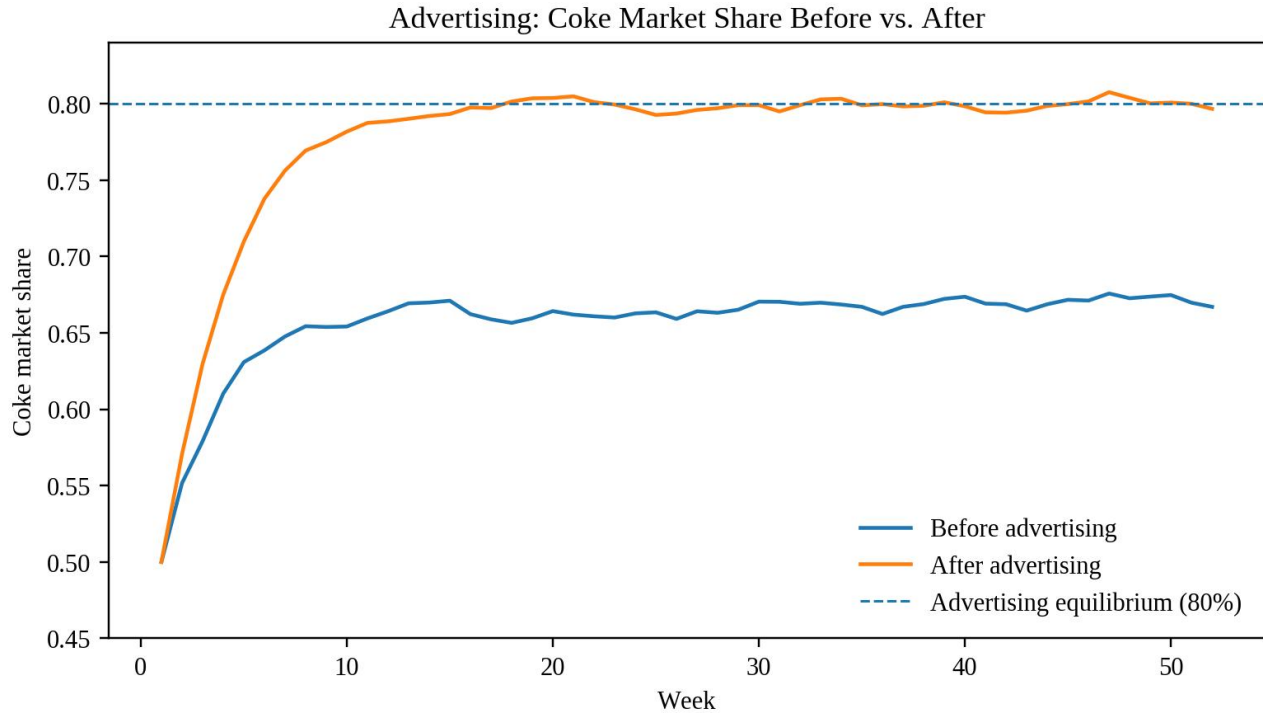


Figure 3. Coke market share before and after advertising. Advertising raises Coke retention from 90% to 95%.

Advertising changes the transition matrix by reducing weekly Coke-to-Pepsi switching from 10% to 5%, while Pepsi-to-Coke switching remains 20%. The implied stationary Coke share under this modified matrix is **80%**. The simulated advertising path ends Week 52 at **79.66%**, close to that analytical benchmark. Its 52-week average Coke share is 77.74%, compared with 65.62% under the baseline path.

The economic test is more demanding than the market-share test. Baseline annual profit is \$341,209. After improved retention and deduction of the \$50,000 annual advertising cost, simulated annual profit is **\$354,243**. The resulting **incremental gain is \$13,034**, equivalent to **3.82%** of baseline annual profit. In this workbook realization, **advertising therefore creates positive net economic value despite its substantial fixed cost**.

4.4 Coupon Experiment

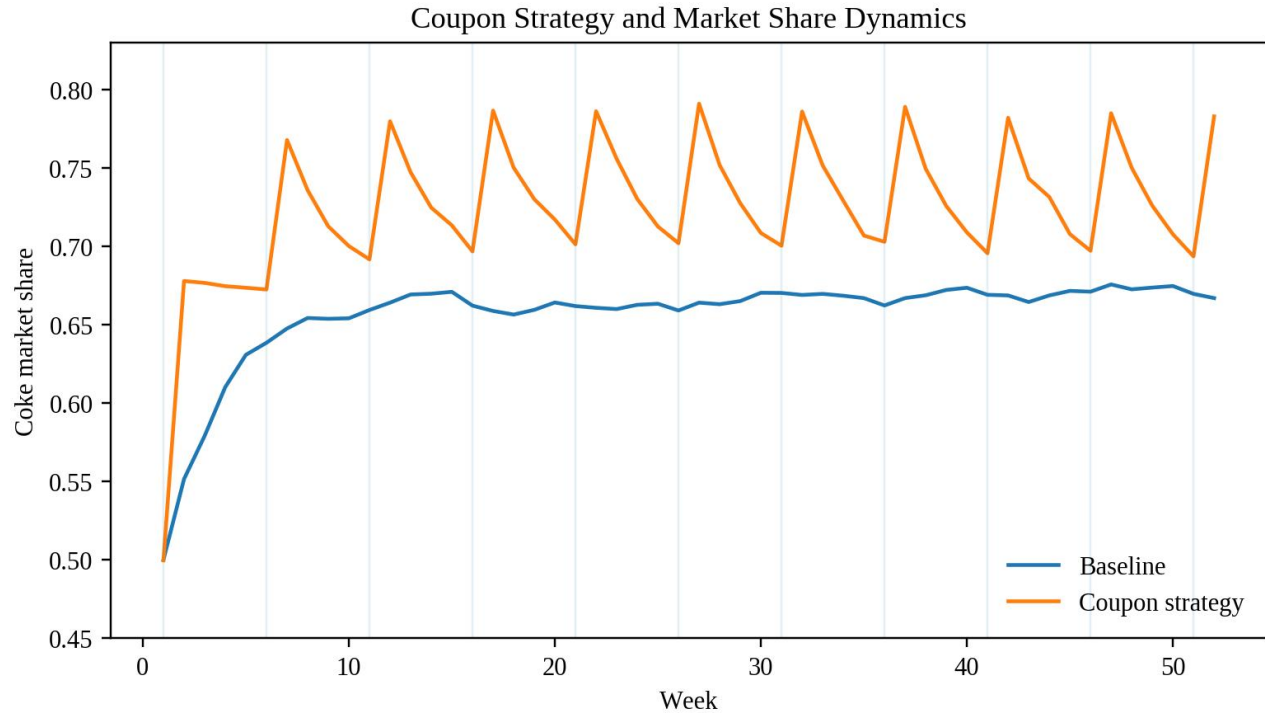


Figure 4. Baseline and coupon-strategy Coke market share. Vertical markers identify coupon weeks used in the workbook.

The **coupon strategy** changes both price and switching behavior. In coupon weeks, Coke retention rises to 95% and Pepsi retention falls to 60%, implying that 40% of current Pepsi customers switch to Coke. At the same time, Coke sacrifices \$0.25 of the \$1.00 per-case profit. The resulting market-share path displays a **repeated intervention-and-decay pattern** rather than convergence to a single stationary level, because the transition matrix changes periodically. Coke share ends Week 52 at **78.29%**, and its average across the simulated year is **72.40%**.

Annual profit with coupons is **\$357,852.75**, compared with \$341,209 under the no-coupon baseline. The **incremental gain is \$16,643.75**, or **4.88%** of baseline profit. Thus, in this realization, **the additional switching toward Coke more than offsets the margin sacrificed in promotion weeks**.

4.5 Strategy-Level Profit Comparison

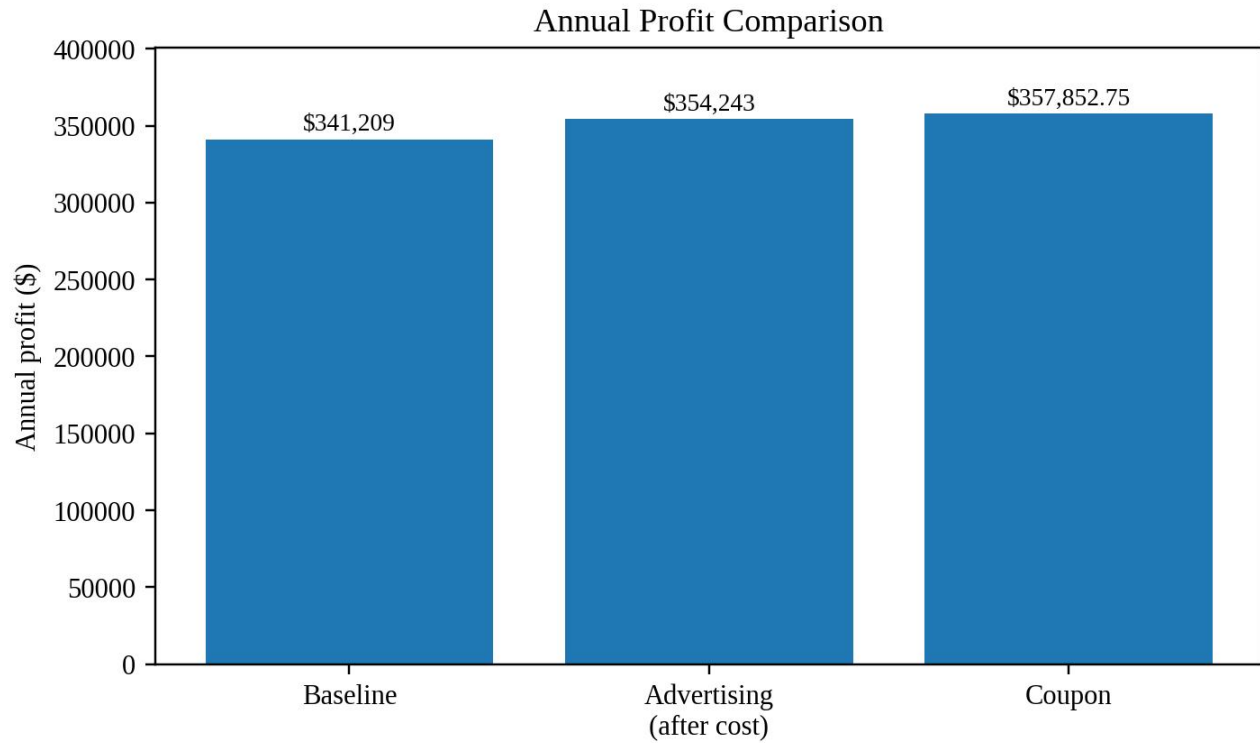


Figure 5. Annual profit comparison from the final workbook outputs.

Strategy	Annual profit	Increment vs. baseline	Increment (%)
Baseline	\$341,209.00	—	—
Advertising (after cost)	\$354,243.00	\$13,034.00	3.82%
Coupon	\$357,852.75	\$16,643.75	4.88%

Both interventions outperform the baseline in the current simulation. The **coupon strategy** produces the **highest** annual profit, exceeding the advertising strategy by \$3,609.75. This is a result of the specific transition effects, costs, discount, and random realization encoded in the workbook; it should not be interpreted as a universal ranking of advertising and coupons outside those assumptions.

5. Interpretation and Model Validation

5.1 Convergence as a Structural Check

The convergence experiment is more than a visual result: it tests whether the simulated process behaves consistently with the theoretical Markov structure. If the transition matrix is fixed, an ergodic two-state chain should progressively lose memory of its initial state. The three initial-share scenarios do exactly that. Their Week-52 shares cluster within less than one tenth of a percentage point of the theoretical equilibrium. This agreement between analytical expectation and stochastic simulation increases confidence that the spreadsheet logic is implementing the intended transition process.

5.2 Cross-Validation Through Micro and Macro Models

The **micro and macro models** provide a second validation layer. The micro implementation generates more than half a million customer-week states and then aggregates them; the macro implementation simulates

customer counts directly. Their Week-52 Coke shares differ by only 0.60 percentage points in the current realization. The small discrepancies across weeks are expected because the two sheets use separate random draws. The substantive agreement indicates that the aggregation logic and transition mechanics are internally coherent.

5.3 Interpreting Stochastic Output

RAND() and BINOM.INV() make the workbook volatile: recalculating the file generates a new sample path. Consequently, a single numerical realization should not be treated as a deterministic forecast. The strongest evidence in this project comes from structural patterns that are expected to persist across **recalculations**: convergence toward the baseline equilibrium, a higher equilibrium region after increasing Coke retention, periodic share increases under coupon weeks, and the need to compare intervention benefits against costs. Simulation validation literature emphasizes that confidence comes from multiple forms of checking—assumptions, transformations, and output behavior—rather than from a single test (Van Horn, 1971).

6. Managerial and Analytical Implications

6.1 Retention Has Dynamic Value

Advertising in this model does not simply generate one additional purchase. It changes the **probability** that a current Coke customer remains a Coke customer next week. Because next week's state becomes the starting point for the following transition, retention improvements propagate forward. This **compounding state effect** explains why a five-percentage-point improvement in Coke retention can move the long-run share benchmark from 66.67% to 80%.

6.2 Market Share Is an Intermediate Metric, Not the Objective

The advertising and coupon experiments illustrate why managers should avoid evaluating marketing programs solely by share lift. Advertising raises share but requires a fixed \$50,000 payment. Coupons raise switching toward Coke but reduce margin in designated weeks. In both cases, the relevant objective is **incremental profit** after accounting for the economic cost of changing customer behavior. This distinction is especially important in dynamic markets, where a strategy can look attractive on volume while destroying value on margin.

6.3 Promotion Creates a Time-Varying Competitive System

The coupon experiment is analytically different from the advertising experiment. **Advertising** produces a new **fixed transition matrix** and therefore a new **stationary equilibrium**. **Coupons** create a **periodic transition process**: the market alternates between normal and promotion-week behavior. The appropriate managerial interpretation is therefore **cyclical** rather than stationary. **Promotion temporarily attracts competitor customers, after which ordinary loyalty and switching probabilities begin to reassert themselves until the next intervention.**

6.4 Decision Use

A practical manager could use this framework as a **scenario engine**. Estimated retention and switching rates can be inserted into the transition matrix; campaign costs, discount depth, and timing can be changed; and the resulting market-share and profit distributions can be compared. The value of the framework is its transparency: customer behavior assumptions are explicit, the path from behavior to share is traceable, and the financial consequence of each intervention is visible.

7. Limitations and Extensions

The model is intentionally parsimonious, which makes the mechanics transparent but also limits external validity. The most important limitations are listed below.

- Homogeneous customers. All Coke customers share the same retention probability and all Pepsi customers share another. Real markets contain heterogeneous loyalty, price sensitivity, and promotion response.
- Two-brand closed market. Entry, exit, outside options, category expansion, and changes in total market size are excluded.
- Fixed baseline probabilities. The model does not estimate transition probabilities from observed panel data, nor does it allow them to drift with seasonality, competitor actions, or macro conditions.
- Intervention effects are assumed rather than causally estimated. The 95% Coke retention under advertising and coupon-week switching rates are scenario inputs, not empirical treatment effects.
- Single realized path. The workbook demonstrates stochastic behavior but does not summarize hundreds or thousands of independent annual replications with confidence intervals for profit or market share.
- Simplified economics. Profit is represented as contribution per case minus the specified advertising cost or coupon discount. Production capacity, channel margins, customer acquisition costs, inventory, and competitive retaliation are outside the model.

The most useful extension would be a repeated-run experiment. Running many independent 52-week simulations would produce distributions for annual profit, final market share, and the incremental value of each strategy. Confidence intervals, probability of loss, and downside-risk measures could then supplement the point estimates. A second extension would introduce customer segments with different transition matrices, consistent with research that models heterogeneity in brand-switching behavior. A third would estimate transition probabilities and intervention effects from transaction or panel data rather than treating them as fixed scenario inputs.

8. Conclusion

This project demonstrates how a simple **stochastic state-transition model** can connect **consumer behavior**, **market-share dynamics**, and **marketing economics**. Under the **baseline** transition matrix, Coke's theoretical equilibrium share is **66.67%**, and both **macro** and **micro** simulations converge near that level. **Changing the initial market share alters the path but not the long-run region. Changing the transition probabilities, however, changes the competitive structure itself.**

In the final workbook realization, **advertising** increases annual profit from \$341,209 to \$354,243 after its \$50,000 cost, while the **coupon** strategy produces **\$357,852.75**. The results show that **retention and switching can have persistent effects because customer states carry forward through time**. They also reinforce a central managerial principle: **market-share improvement is valuable only when the dynamic economic benefit exceeds the cost required to create it.**

Methodologically, **the project combines an analytical equilibrium benchmark, aggregate stochastic simulation, individual-level simulation, scenario analysis, and profit evaluation in one transparent Excel model.** That combination makes the workbook useful not only as a classroom simulation, but as a compact demonstration of how **probabilistic modeling** can support business decision analysis.

References

- Bielza, C., Müller, P., & Ríos Insua, D. (1999). Decision analysis by augmented probability simulation. *Management Science*, 45(7), 995–1007. <https://doi.org/10.1287/mnsc.45.7.995>
- Givon, M., & Horsky, D. (1994). Intertemporal aggregation of heterogeneous consumers. *European Journal of Operational Research*, 76(2), 273–282. [https://doi.org/10.1016/0377-2217\(94\)90107-4](https://doi.org/10.1016/0377-2217(94)90107-4)
- Grover, R., & Dillon, W. R. (1988). Understanding market characteristics from aggregated brand switching data by the method of spectral decomposition. *International Journal of Research in Marketing*, 5(2), 77–89. [https://doi.org/10.1016/0167-8116\(88\)90061-4](https://doi.org/10.1016/0167-8116(88)90061-4)
- Leong, T.-Y. (2007). Monte Carlo spreadsheet simulation using resampling. *INFORMS Transactions on Education*, 7(3), 188–198. <https://doi.org/10.1287/ited.7.3.188>
- Van Horn, R. L. (1971). Validation of simulation results. *Management Science*, 17(5), 247–258. <https://doi.org/10.1287/mnsc.17.5.247>

Appendix A. Workbook Architecture

Worksheet	Role
00_Project_Overview	Portfolio-facing project narrative: problem, model, design, interpretation, implications, and limitations.
1_MarketShare_Macro	Aggregate 52-week simulation; initial-share, advertising, and coupon experiments.
1_MarketShare_Micro	10,000-customer individual simulation and aggregation to weekly market share.
01_Results_Interpretation	Linked KPI summary, convergence outputs, strategy interpretation, and visual comparisons.

The report uses the final workbook's cached simulation outputs as the numerical record. Because the workbook contains volatile random functions, recalculating Excel can produce different realized values while preserving the same model structure.